

# Developing Interactive Instructional Resources for STA258 Using Quarto

**Statistics with Applied Probability**

**Sarah Halime & Ashwin Selvapandian**

Department of Mathematical and Computational Sciences  
University of Toronto Mississauga

- 1 Motivation and goals
- 2 Building the resource
- 3 From concept to classroom interaction
- 4 Development and refinement
- 5 Limitations and future work
- 6 Acknowledgements



## Scan to explore the course resource

- Chapter landing pages
- Interactive Reveal.js slide decks
- WebR code blocks and exercises
- PDF views for static access

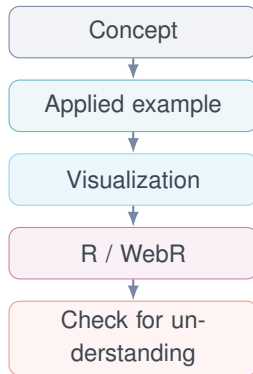


[https://nishanmudalige.github.io/  
STA258\\_Slides/](https://nishanmudalige.github.io/STA258_Slides/)

# Why this project?

A statistics course asks students to move repeatedly between:

- mathematical ideas,
- interpretation in context,
- visualization,
- computation in R, and
- practice and feedback.



**Project aim:** bring these steps into one coherent, browser-based learning experience.

**Clarity** Keep statistical reasoning visible with concise explanations and deliberate visual hierarchy.

**Interactivity** Let students calculate, visualize, run R, and check understanding directly inside the slides.

**Consistency** Use a common structure, visual language, and interaction pattern across the course.

**Applied context** Anchor procedures in realistic examples before formal computation.

**Reproducibility** Keep source, computation, styling, and publishing within a maintainable workflow.

**Accessibility** Reduce software barriers by moving core interactions into the browser.

The design goal was not to make slides “more interactive” in isolation, but to make the learning sequence more coherent.

## Chapter-based course structure

Chapter landing pages and browser-based slide decks organized across the STA258 course sequence.

## Statistical visualization

Reproducible R graphics, Plotly figures, and selected JavaScript widgets designed around applied examples.

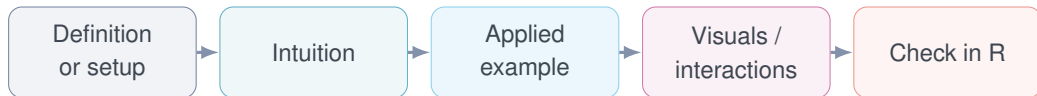
## Interactive computation

WebR, R demonstrations, solution reveals, and quiz interactions embedded directly within the learning materials.

## A maintainable publishing system

Shared CSS, Git/GitHub workflow, rendered HTML/PDF access, and documentation for future maintainers.

# The learning sequence



- Repeating the same structure gives students a **predictable rhythm** from one topic to the next.
- Interactive elements appear where students can apply or check their understanding of the statistical idea.

- **Quarto** for authoring and rendering
- **Reveal.js** for browser-based presentations
- **R / knitr** for reproducible calculations and figures
- **WebR** for live R execution in the browser
- **Plotly** and JavaScript for interactive visualizations
- **HTML/CSS** for custom layouts and interaction design
- **Quiz plugins** for embedded formative checks
- **Git/GitHub** for collaborative version control
- **GitHub Pages** for deployment
- **PDF views** for printable / static access

The result is one workflow from:

**source code** → **computation** → **interaction** → **published course resource**

## Run R

WebR lets students execute selected R code without leaving the slide deck or installing software locally.

## Reveal thinking

Fragment order controls when setup, calculation, and result appear during instruction.

## Immediate checks

Embedded quizzes and solution reveals provide quick formative feedback.

**Interaction principle:** each click should move the statistical reasoning forward.

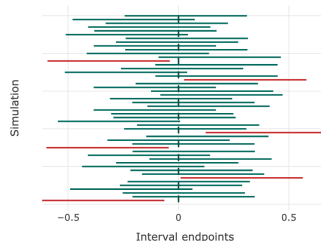
## 5.2 Interpreting Confidence

For a  $C\%$  confidence interval:

In repeated sampling, about  $C\%$  of intervals constructed using this method will capture the true parameter.

Do not say: “There is a 95% probability that this particular interval contains  $\mu$ .”

The parameter is fixed. The random object is the interval produced by the sample.



Chapter 5 — interpreting confidence intervals, with a simulated-coverage Plotly figure

- The resource moves from **context** to **visual reasoning** before or alongside computation.

# Example 2: Check in R

## Example: Active Minutes

A smartwatch logs **daily active minutes** (time spent moving) for a random sample of 15 people. Active minutes are normally distributed with known  $\sigma = 15$  minutes:

79, 43, 58, 66, 101, 63, 79, 33, 58, 71, 60, 101, 74, 55, 88

Construct a 95% CI for  $\mu$ .

R Code

```
1
2 minutes <- c(79,43,58,66,101,63,79,33,58,71,60,101,74,55,88)
3 xbar <- mean(minutes)
4 n <- length(minutes)
5 sigma <- 15
6 zstar <- qnorm(0.975)
7 ci <- xbar + c(-1, 1) * zstar * sigma/sqrt(n)
8
9 xbar
10 ci
```

$\bar{x} = 68.6$ , so the interval is approximately (61.01, 76.19). We are 95% confident that the population mean number of daily active minutes is between about 61 and 76.

Chapter 5 — a worked example with a live WebR editor beside it

- Connects the hand-worked statistical method directly to its implementation in R.
- Especially valuable because STA258 tutorials require R, and this support carries into upper-year statistics courses.

## Example 3: Formative feedback

**Question**

A sample has  $\bar{x} = 42$ ,  $n = 36$ , and known  $\sigma = 12$ . For a 95% CI, what is the margin of error?

4.00

1.96

7.84

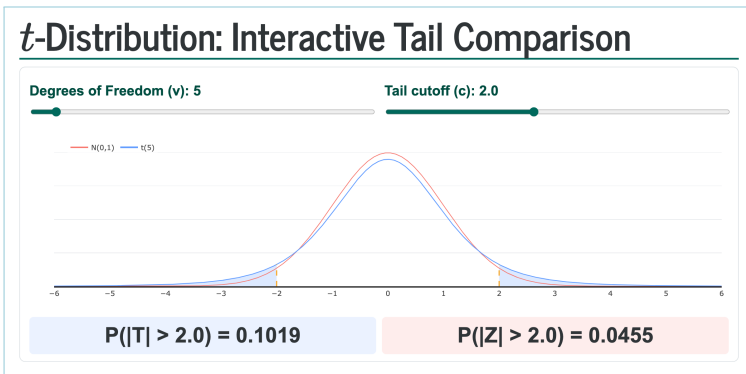
3.92

[Check](#) [Reset](#) [Prev](#) [Next](#)

Chapter 5 — an embedded quiz with immediate checking and explanations

- The goal is not simply to reveal an answer, but to make the reasoning and common misconception visible.

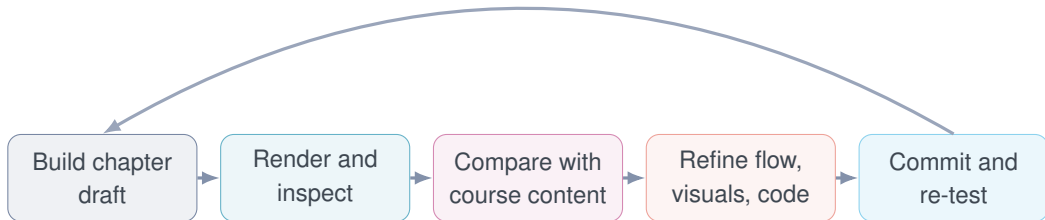
# Example 4: Interactive statistical visualization



Chapter 4 — a  $t$ -distribution tail explorer with live sliders for  $\nu$  and the cutoff

- The interaction reinforces what changes, what stays fixed, and what the graphical objects represent statistically.

# An iterative summer development process



- Refinement focused on **genuine instructional discrepancies**: incorrect formulas, broken interaction order, misleading wording, inconsistent examples, and serious layout issues.
- The workflow also standardized chapters without changing content simply for cosmetic uniformity.

## Instructional consistency

- Example flow and fragment order
- Notation and final-answer precision
- R checks that match the worked example
- Conclusions stated in context
- Chapter-to-chapter navigation

## Visual / technical consistency

- Fixed scaling and overflow
- Consistent Plotly presentation
- Readable axes and labels
- Solution / reset behavior
- Shared CSS and reusable layouts

Consistency became a **course-level design problem**, not just a collection of isolated slide edits.

# Designed for maintainability

- Shared styling is centralized rather than duplicated across slides.
- Chapter files follow a predictable structure.
- A help manual documents rendering, navigation, WebR, quiz behavior, Git workflow, and common errors.
- The repository structure separates editable source from rendered output.

`chapterN-slides.qmd`

`shared styles.css`

`_extensions/ + WebR`

HELP / maintenance guide

`rendered docs/`

# What the project enables

## For students

A single browser-based path connecting explanation, visualization, computation, and formative practice.

## For instructors

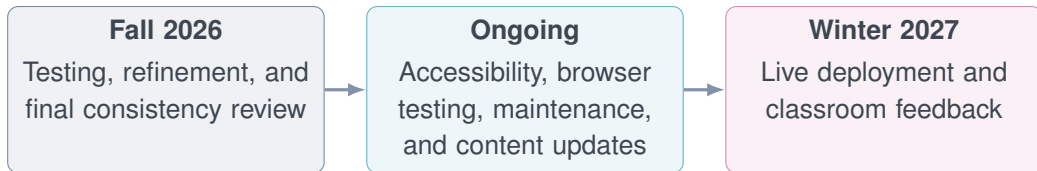
Reusable teaching sequences with controlled reveal order, consistent examples, and integrated R demonstrations.

## For future developers

A documented structure that can be updated chapter-by-chapter without rebuilding the entire system.

The contribution is both **instructional** and **infrastructural**: the content and the system used to deliver it are developed together.

- The resource is still in development and has not yet been fully evaluated through live course deployment.
- Browser-based interactivity introduces dependencies on rendering, JavaScript, and WebR behavior across devices.
- Interactive slides require more maintenance than static lecture material because code, styling, and pedagogy must remain synchronized.
- Accessibility and mobile responsiveness require continued testing as the resource evolves.



- Use classroom feedback to determine which interactions genuinely support learning and which should be simplified.
- Continue developing the resource as a maintainable course platform rather than a one-time slide deck.

1. **Quarto provides a single authoring environment** for text, mathematics, R, HTML/CSS, and interactive presentation components.
2. **Interactivity is most useful when it follows the statistical reasoning**, rather than being added as a separate feature.
3. **Consistency across chapters matters pedagogically**: students can focus on the concept when the structure and interaction patterns are predictable.
4. **Maintainability is part of instructional design**: future course updates should be possible without reconstructing the system.

We would like to thank:

- Professor Nishan Mudalige, for his guidance, encouragement, and thoughtful feedback throughout the development of this project;
- the Department of Mathematical and Computational Sciences at the University of Toronto Mississauga.

- Mudalige, N., Halime, S., & Selvapandian, A. (2026). *STA258: Statistics with Applied Probability — Interactive instructional resources*. University of Toronto Mississauga. [https://nishanmudalige.github.io/STA258\\_Slides/](https://nishanmudalige.github.io/STA258_Slides/)
- Alili, N., & Su, X. (2025). Student-faculty co-creation of open educational resources for learning applied statistics with open source software tools. *Journal of Computing, Data, and Exploration*, 1(1). <https://doi.org/10.33137/codex.v1i1.45881>
- American Statistical Association Revision Committee. (2016). *Guidelines for Assessment and Instruction in Statistics Education (GAISE) College Report 2016*. American Statistical Association.  
[https://www.amstat.org/asa/files/pdfs/gaise/gaisecollege\\_full.pdf](https://www.amstat.org/asa/files/pdfs/gaise/gaisecollege_full.pdf)
- Posit. (n.d.). *Quarto documentation*. <https://quarto.org/docs/>
- R-universe / WebR Project. (n.d.). *WebR: R in the browser*. <https://docs.r-wasm.org/webr/latest/>
- R Foundation. (n.d.). *R: The R Project for Statistical Computing*. <https://www.r-project.org/>
- University of Toronto Mississauga. (n.d.). *STA258H5: Statistics with Applied Probability*. Academic Calendar.  
<https://utm.calendar.utoronto.ca/course/sta258h5>

# Thank you!

Questions or comments?

[https://nishanmudalige.github.io/STA258\\_Slides/](https://nishanmudalige.github.io/STA258_Slides/)

